Sparse optimisation and quantum-inspired encoding for ransomware detection

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Abstract. Ransomware remains a persistent cybersecurity threat difficult to detect due to high-dimensional network traffic and sophisticated obfuscation techniques. Existing feature selection methods often struggle with redundancy, noise, and the curse of dimensionality, leading to poor generalisation and limited interpretability in ransomware detection. To address these challenges, we propose BioSparse-MCP, a hybrid feature selection framework that integrates gradient-based optimisation with the Minimax Concave Penalty (MCP) to enforce sparsity, alongside a Rotated Circular Partitioning (RCP) strategy to improve the spatial organisation of selected features. This design reduces redundancy, enhances discriminative power, and provides rotation-aware representations that overcome the limitations of conventional dimensionality reduction. The framework further incorporates a Quantum Feature Mapping (QFM)-inspired geometric transformation, in which features are projected onto a spherical space, rotated, and partitioned into angular sectors, while preserving linear computational complexity. All RCP and QFM operations are classically simulated, ensuring compatibility with conventional machine learning pipelines and real-time deployment without specialised hardware. Implemented in Python using standard numerical libraries, BioSparse-MCP was evaluated on 149,043 network traffic instances with an ensemble of KNN and LSTM models. The approach achieved high detection accuracy with a low False Positive Rate (0.25%). Feature attribution analysis highlights cryptocurrency addresses, threat signatures, and IP-level features as key contributors. These results demonstrate that combining sparse optimisation with quantum-inspired geometric encoding provides an efficient and interpretable solution for ransomware detection in high-dimensional network environments.

Keywords: Ransomware Detection · Machine Learning · Quantum Feature Mapping · Sparse Optimisation · Minimax Concave Penalty · Rotated Circular Partitioning

1 Introduction

Cybercrime has emerged as an unavoidable consequence of digital transformation with global damages projected to reach 10.5$ trillion by 2025 [1]. Among the most prominent threats is ransomware, a form of malicious software that encrypts victims' data and demands a ransom for its release. The implications are particularly severe when ransomware targets critical infrastructures where disruptions can jeopardise not just economic stability but also human lives [2], [3]. Despite increasing awareness, ransomware continues to penetrate organisations at alarming rates. Fortinet's 2023 Global Ransomware Research and Sophos's 2024 report revealed a gap between perceived readiness and actual resilience. Ransomware detection techniques have advanced into static analysis, which inspects file properties, and dynamic analysis, which observes behavior during runtime [4]. Dynamic analysis often used in conjunction with static methods to form hybrid detection systems has gained importance [5], [3]. Given their capacity to capture behavioral patterns during execution, dynamic features offer a rich foundation for ransomware detection. Among these, Application Programming Interface (API) calls are the most utilised due to their presence across all system level operations, especially those involving kernel activity. Additional features such as registry keys, file system interactions, Input or Output (I/O), and process memory operations provide complementary behavioral indicators [1], [6]. Network traffic analysis is also common, particularly for identifying Command and Control (C&C) anomalies. Nevertheless, significant issues persist, like the absence of standardised, and high-quality datasets. Approximately 80% of studies address this by creating custom datasets, by executing ransomware samples in controlled environments [1]. Unfortunately, poor labeling practices, lack of feature extraction transparency, and inconsistent methodologies undermine reproducibility and hinder real-world applicability [7], [8]. Motivated by these gaps, this study proposes a novel feature selection algorithm (*BioSparse-MCP*) designed to improve the detection of ransomware attacks in high dimensional datasets. These datasets often suffer from redundancy, noise, and the curse of dimensionality, which hinder model training and predictive performance. In the context of ransomware detection, raw datasets derived from network traffic can contain thousands of features, many of which are weakly informative. Recent advances in feature selection methods have enabled more efficient data handling by retaining only the most discriminative features, reducing computational complexity and enhancing classification accuracy. Building on these advancements, feature selection proves essential in mitigating overfitting, improving generalisation, and accelerating training on high dimensional data. To address these challenges, this study introduces BioSparse-MCP, a biologically inspired algorithm that combines gradient descent with the *Minimax Concave Penalty (MCP)* regularisation. It also integrates a quantum inspired module called *Rotated Circular Partitioning* (RCP) for enhanced ransomware classification. The design of BioSparse-MCP draws on neuroscience and evolutionary biology concepts, including weight shrinkage, selective neuron retention, evolutionary adaptation, pruning mechanisms, survival of the fittest, cortical column organisation, and stimulus encoding spaces. These principles guide the dimensionality reduction process, making the algorithm highly suited for complex cybersecurity datasets. The classification framework incorporates an ensemble of *K-Nearest Neighbor (KNN)* and *Long Short-Term Memory (LSTM)* models to assess BioSparse-MCP's effectiveness. To the best of the authors' knowledge, this is the first work to apply BioSparse-MCP to the network traffic dataset used in [9] for ransomware detection. The main contributions of this study can be summarised as follows:

1. **BioSparse-MCP Algorithm:** We introduce a biologically inspired feature selection framework that integrates gradient descent with the MCP. This approach enforces sparsity while retaining highly discriminative features, thereby improving interpretability and reducing computational overhead in high-dimensional ransomware datasets.
2. **Quantum-Inspired RCP:** We propose a novel geometric encoding strategy that projects selected features onto a spherical manifold, applies rotational transformations, and partitions them into angular sectors. This quantum-inspired design enhances feature disentanglement and provides rotation-aware representations, while remaining fully classical and hardware-independent.
3. **Clarification of Quantum-Inspired Learning:** The study explicitly distinguishes quantum-inspired encodings (implemented on classical architectures) from Quantum Machine Learning (QML) approaches that rely on genuine quantum circuits or hardware, ensuring conceptual clarity for readers and practitioners.
4. **Ensemble Classifier:** We design and implement a hybrid ensemble model that combines the geometric strengths of K-Nearest Neighbors (KNN) with the temporal modeling capacity of Long Short-Term Memory (LSTM). This ensemble achieves high detection accuracy and a low false positive of 0.25%, demonstrating the effectiveness of combining sparse optimisation with quantum-inspired encoding.
5. **Empirical Validation:** The proposed framework is evaluated on 149,043 network traffic instances, highlighting the importance of cryptocurrency addresses, and threat signatures features in ransomware detection. The results confirm that BioSparse-MCP provides an efficient and interpretable solution for real-time cybersecurity environments.

Together, these contributions advance the state of the art in ransomware detection by bridging biologically inspired optimisation with quantum-inspired geometric encoding, while maintaining compatibility with conventional pipelines. Unlike Su's model [9], which uses the Chi-Square method to reduce dimensionality by eliminating irrelevant features prior to training, BioSparse-MCP functions as an embedded regularisation technique. The optimisation process employs a *proximal gradient descent* algorithm that leverages the proximal operator of MCP to iteratively update and prune feature weights in a biologically inspired manner. Prior to optimisation, the *Integrated Gradients (IG)* method is applied to identify the most informative samples by computing feature attributions. This guides the algorithm toward salient regions in the input space. Finally, the quantum-inspired *RCP module* performs geometric post-partitioning of selected features, enabling robust, rotation-aware representation of ransomware behavior. Therefore, this research is guided by the central question: *How can quantum feature encoding and selection improve the detection of ransomware within network traffic datasets*? The remainder of this paper is organised as follows: Section 2 discusses related works. Section 3 introduces the proposed framework. Section 4 discusses the results. Finally, Section 5 concludes.

2 Related Works

Recent research on ransomware detection has explored diverse Machine Learning (ML) approaches and feature selection strategies. Rios et al. [1] proposed a classification framework showing that Random Forest (RF) achieved perfect offline accuracy, while Multi-Layer Perceptron (MLP) generalised better in real-world scenarios. Dutta and Karmakar [10] demonstrated that AdaBoost outperformed Support Vectors Machine (SVM) on network traffic data, highlighting the strength of ensemble methods. Dimensionality reduction has also been a focus. Mohamed et al. [4] introduced an Adaptive WavePCA-Autoencoder with meta-attention, while Zahoora et al. [11] employed a contractive autoencoder to detect both known and unknown ransomware. Wang et al. [12] utilised an LSTM encoder-decoder to capture behavioral patterns, achieving notable accuracy gains. Explainability has been emphasised by Jalil [13], who integrated interpretable Artificial Intelligence (AI) into ransomware detection frameworks. Hasan et al. [14] systematically compared feature selection techniques (none, Linear Discriminant Analysis (LDA), Principal Component Analysis (PCA)) across twelve algorithms, showing that ensemble methods paired with PCA consistently outperformed traditional models. Across these works, a recurring gap lies in the limited application of advanced feature selection methods for high-dimensional cybersecurity datasets. Distinct from prior approaches, the proposed algorithm (*BioSparse-MCP*) is biologically inspired, drawing on principles such as survival of the fittest and sparse activation. And integrates quantum-inspired feature encoding to enhance precision and dimensionality reduction. This hybrid design leverages gradient descent with MCP regularisation, while RCP introduces angular structure into the feature space. Together, these innovations enable more robust and interpretable ransomware detection compared to conventional dimensionality reduction or ensemble learning strategies (Figure 1).

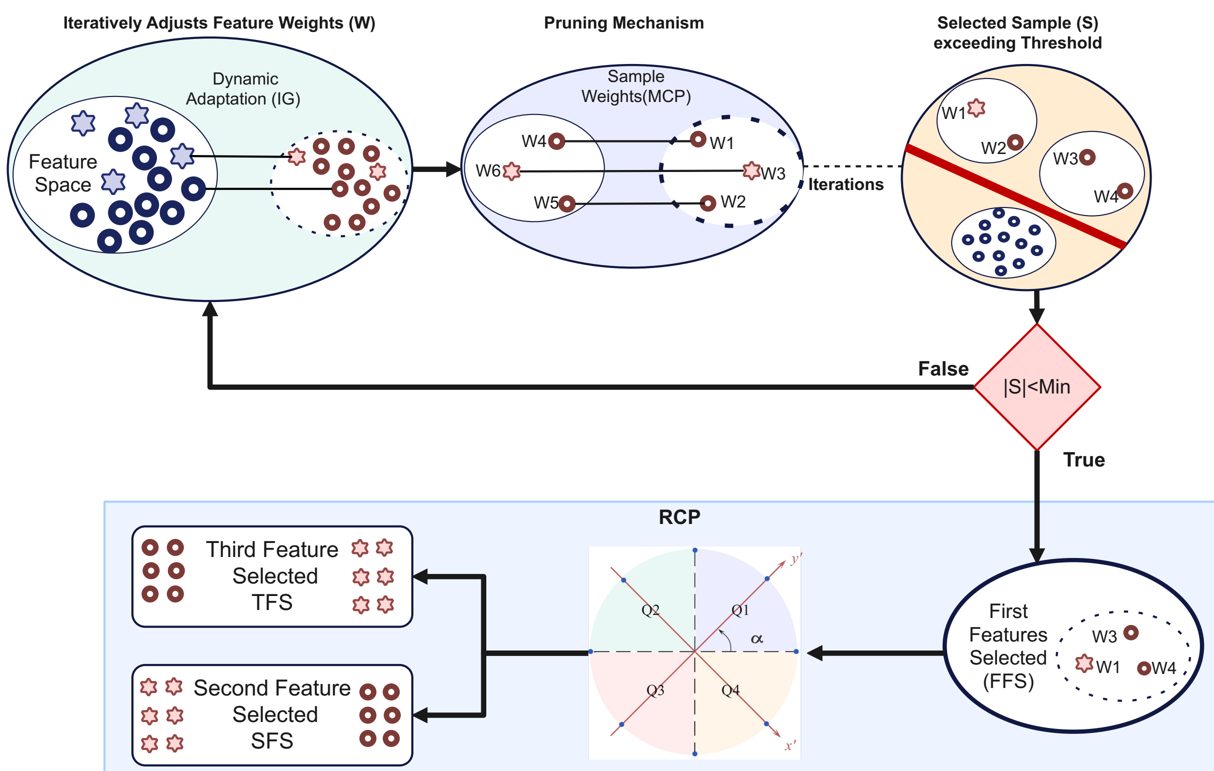


Fig. 1: Complete flow diagram of the proposed feature selection technique.

3 The Framework Description

The proposed framework is shown in Figure 1. The study uses the UGRansome dataset (`https://www.kaggle.com/datasets/nkongolo/ugransome-dataset/data`), a recent dataset for ransomware classification and Intrusion Detection System (IDS) evaluation, containing 149,043 records of normal and malicious traffic [9]. The dataset captures ransomware communication patterns, providing network flow features such as bytes transferred, malicious addresses, timestamps, and protocol flags (Table 1). It presents a balanced distribution of traffic types, with approximately 46.5% representing benign activity and 53.5% attributed to ransomware (Table 2). During the data cleaning process, duplicate was removed and the final selected feature set resulted in 128,262 records. The binary classification of the target variable (ransomware and benign) enhances the dataset's suitability for supervised ML tasks aimed at distinguishing ransomware activity from legitimate network behavior (Figure 1). The UGRansome dataset has been previously utilised in studies by Yan et al. [15], Mohamed et al. [4], Ochoa et al. [1], Chaudhary and Adhikari [16], Su [9], and Sokhonn et al. [17].

Table 1: Description of the UGRansome Dataset

Feature	**Description**	**Encoding**	**RZ**	**Selected?**
`Time`	Timestamp of the network flow	q_0	0.22	✓
`Protocol`	Network protocol	q_1	0.44	✓
`Flag`	Session control signals	q_2	0.12	×
`Family`	Ransomware family name	q_3	0.84	✓
`Clusters`	Behavioral cluster label of malware	q_4	0.00	×
`SeedAddress`	Cryptocurrency wallet address	q_5	0.47	✓
`ExpAddress`	Additional address	q_6	0.68	✓
`BTC`	Ransom demand in Bitcoin (BTC)	q_7	0.54	✓
`USD`	Ransom demand in Dollars	q_8	0.40	✓
`Netflow`	Total bytes transferred in network flow	q_9	0.61	✓
`IPaddress`	Source or destination address	q_{10}	0.61	✓
`Threats`	Known threats or malware signatures	q_{11}	0.58	✓
`Port`	Network port number	q_{12}	0.14	✓
`Prediction`	Target (1: ransomware, 0: benign)	q_{13}	0.47	✓

Table 2: UGRansome Benign and Ransomware Traffic

Traffic Type	**Category**	**%**
User Datagram Protocol (UDP)	Benign	12.9
Secure Shell (SSH)	Benign	16.8
Scanning	Benign	7.9
Ports	Benign	8.9
Total Benign		**46.5**
Blacklist	Ransomware	13.9
Botnet	Ransomware	11.9
Denial-of-Service (DoS)	Ransomware	9.9
Nerisbotnet	Ransomware	3.0
Spam	Ransomware	14.9
Total Ransomware		**53.5**

3.1 Proposed feature selection

BioSparse-MCP iteratively adjusts feature weights vector $\mathbf{w}$ using proximal gradient descent [18], reflecting the dynamic neural adaptation observed in biological systems. At each iteration t, given a learning rate η, the forward update is computed in Equation 1.

$$\mathbf{w}_{t+\frac{1}{2}} = \mathbf{w}_t - \eta \nabla \mathcal{L}(\mathbf{w}_t), \tag{1}$$

where $\mathcal{L}(\mathbf{w}) = \frac{1}{n}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$ denotes the Mean Squared Error (MSE). The algorithm then applies the MCP proximal operator [19], which acts as a pruning mechanism (Equation 2).

$$\mathbf{w}_{t+1} = \mathrm{prox}_{\eta\lambda \times \mathrm{MCP}}(\mathbf{w}_{t+\frac{1}{2}}), \tag{2}$$

with the coordinate-wise update given in Equation 3.

$$\mathrm{prox}_{\eta\lambda\times\mathrm{MCP}}(w_i) = \begin{cases} \frac{\mathrm{sign}(w_i)(|w_i|-\eta\lambda)}{1-1/\gamma}, & \text{if } |w_i| \le \gamma\lambda, \\ w_i, & \text{if } |w_i| > \gamma\lambda. \end{cases} \tag{3}$$

This operator enforces sparsity by deactivating less relevant features, akin to neuronal pruning, while preserving strong weights, an analogy to survival of dominant neural pathways (Figure 1). Unlike convex regularizers such as Ridge [20], which merely shrink weights, the MCP penalty allows selective retention of informative features with minimal bias. After T iterations, the algorithm selects features exceeding a threshold τ ensuring that only the most significant features survive, analogous to evolutionary selection pressure (Figure 1). If $|\mathcal{S}| < s_{\min}$, the algorithm retains the top $s_{\min}$ weights to guarantee robustness in low-signal scenarios. Throughout training, BioSparse-MCP tracks both loss $\mathcal{L}(\mathbf{w}_t)$ and weight evolution $\mathbf{w}_t$, enabling detailed interpretability of learning dynamics and convergence. Let $\mathbf{X} \in \mathbb{R}^{n\times d}$ denote the input feature matrix and $\mathbf{y} \in \mathbb{R}^n$ the target vector. The objective is to minimize the regularized loss function shown in Equation 4.

$$\min_{\mathbf{w}\in\mathbb{R}^d} \mathcal{L}(\mathbf{w}) + \sum_{j=1}^{d} \rho_{\lambda,\gamma}(w_j), \tag{4}$$

where the empirical loss is the MSE (Equation 5):

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2, \tag{5}$$

and the MCP penalty $\rho_{\lambda,\gamma}$ is defined in Equation 6.

$$\rho_{\lambda,\gamma}(w_j) = \begin{cases} \lambda|w_j| - \frac{w_j^2}{2\gamma}, & \text{if } |w_j| \le \gamma\lambda, \\ \frac{1}{2}\gamma\lambda^2, & \text{otherwise.} \end{cases} \tag{6}$$

Optimisation Procedure. BioSparse-MCP initialises weights $\mathbf{w}^{(0)} \sim \mathcal{U}(0,1)^d$. For each iteration $t = 0, 1, \ldots, T-1$, it performs:

$$\text{Gradient step:} \quad \mathbf{w}^{(t+\frac{1}{2})} = \mathbf{w}^{(t)} - \eta\nabla\mathcal{L}(\mathbf{w}^{(t)}), \tag{7}$$

$$\text{with } \nabla\mathcal{L}(\mathbf{w}) = \frac{1}{n}\mathbf{X}^\top(\mathbf{X}\mathbf{w} - \mathbf{y}), \tag{8}$$

followed by the element-wise MCP proximal update illustrated in Equation 9.

$$w_j^{(t+1)} = \begin{cases} \dfrac{\mathrm{sign}(w_j^{(t+\frac{1}{2})})\left(|w_j^{(t+\frac{1}{2})}| - \eta\lambda\right)}{1-\frac{1}{\gamma}}, & \text{if } |w_j^{(t+\frac{1}{2})}| \le \gamma\lambda, \\ w_j^{(t+\frac{1}{2})}, & \text{otherwise,} \end{cases} \tag{9}$$

for all $j = 1, \ldots, d$.

Feature Selection Rule. At convergence ($t = T$), the algorithm defines selected feature sets (Equation 10).

$$\mathcal{S} = \left\{ j \in \{1, \ldots, d\} \;\middle|\; |w_j^{(T)}| \geq \tau \right\}. \tag{10}$$

If $|\mathcal{S}| < s_{\min}$, update $\mathcal{S}$ to include the $s_{\min}$ indices corresponding to the largest $|w_j^{(T)}|$ values.

Output. The algorithm returns the final weight vector and selected feature indices $\boxed{\left(\mathbf{w}^{(T)},\, \mathcal{S}\right)}$ (Figure 2).

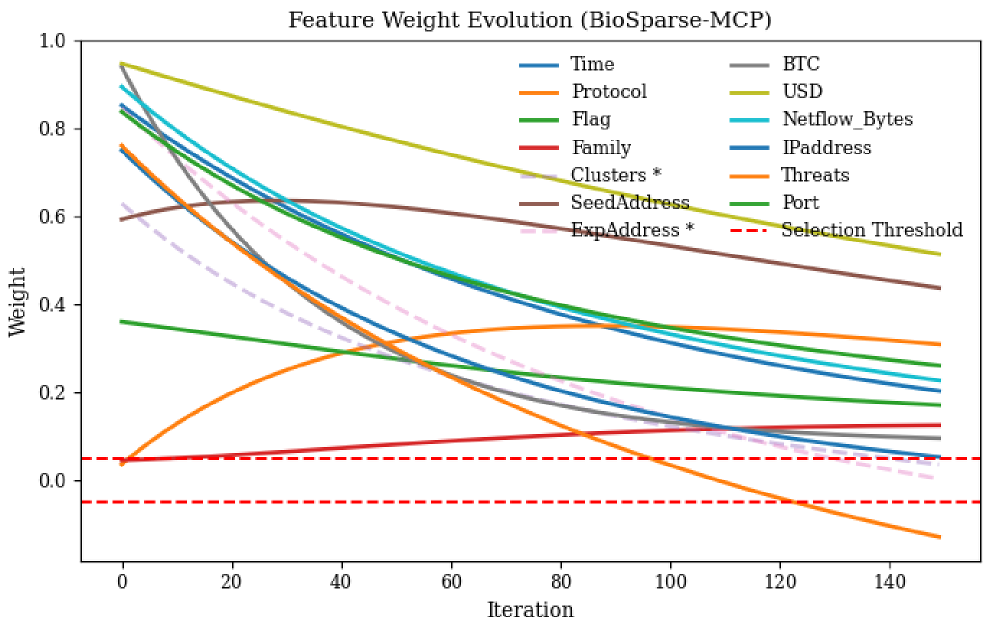


Fig. 2: Feature weight evolution. Irrelevant features are marked with *. For example q_4 for clusters in Table 1.

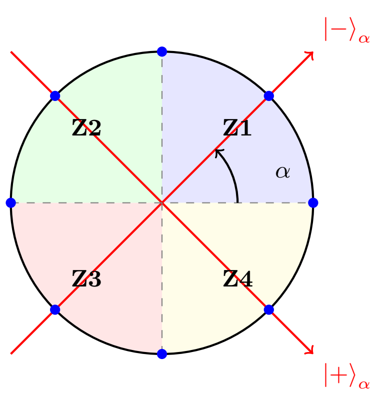


Fig. 3: Quantum inspired feature embedding on the equatorial plane of the Bloch sphere. Features are rotated by angle α and partitioned into four measurement zones (Z1–Z4) based on their projected orientation.

A novel approach to feature partitioning is introduced through the use of a *quantum inspired feature encoding and measurement process*, which geometrically embeds selected features onto the Bloch sphere's equatorial plane [21] and partitions them through simulated quantum measurement outcomes governed by a rotation operator (Figure 3). This quantum process draws strong parallels to how quantum states are prepared and measured, specifically utilising rotation gates and projective measurements to extract feature alignment and orientation [22]. In this analogy, each classical feature is represented as a pure quantum state on the equatorial circle of the Bloch sphere. The feature embedding process maps selected features onto unit vectors parameterized by phase angles, while rotation gates (e.g., $R_z(\alpha)$) simulate angular variation across the feature space.

This process is inspired by neurobiological models of angular encoding and orientation tuning in the primary visual cortex, where stimuli are organised across feature columns in a pinwheel-like structure. For a set of n features $\{f_1, f_2, \ldots, f_n\}$, each feature is encoded into a quantum state using phase encoding, where angular coordinates are defined as $\theta_i = \frac{2\pi i}{n}$, yielding the embedding $(x_i, y_i) = (\cos\theta_i, \sin\theta_i)$. To simulate a rotation gate applied before measurement, a global angle $\alpha \in [0, 2\pi]$ is used, resulting in rotated coordinates via the rotation matrix in Equation 11.

$$\begin{pmatrix} x_i' \\ y_i' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\theta_i \\ \sin\theta_i \end{pmatrix} \tag{11}$$

Post-rotation, each feature undergoes a simulated projective measurement, with its classification into zones determined by the sign of (x_i', y_i'):

$$\text{Zone}(f_i) = \begin{cases} \text{Z1} & \text{if } x_i' > 0 \text{ and } y_i' > 0 \\ \text{Z2} & \text{if } x_i' < 0 \text{ and } y_i' > 0 \\ \text{Z3} & \text{if } x_i' < 0 \text{ and } y_i' < 0 \\ \text{Z4} & \text{if } x_i' > 0 \text{ and } y_i' < 0 \end{cases} \tag{12}$$

This measurement process can be extended to $k \in \mathbb{N}, k \geq 2$ zones by dividing the equatorial plane into k-equal angular slices, analogous to multi-qubit measurement bases in quantum state discrimination. The result is a flexible, rotation-invariant, and quantum-aligned feature partitioning technique inspired by both quantum encoding schemes and neurobiological orientation maps [21], [23].

Complexity. The first phase of the pipeline, involving BioSparse-MCP optimisation, requires computing gradients via $\mathbf{X}^\top(\mathbf{X}\mathbf{w}-\mathbf{y})$ in $\mathcal{O}(nd)$ and applying coordinate-wise MCP updates in $\mathcal{O}(d)$ per iteration, leading to a total complexity of $\mathcal{O}(Tnd)$ over T iterations. Feature selection by thresholding adds at most $\mathcal{O}(d\log d)$, which is negligible. The second phase performs Quantum Feature Mapping (QFM) [24] by embedding the n selected classical features onto a circular layout [23]. Each feature is assigned a rotation and placed into one of the defined sectors, all operations executed in $\mathcal{O}(n)$ time. When r candidate rotations are considered for optimisation, the complexity of this phase increases to $\mathcal{O}(rn)$. Thus, the total computational complexity of the entire pipeline becomes $\boxed{\mathcal{O}(n(Td + r))}$, where d is the original input dimensionality and T denotes the number of iterations used during feature selection or encoding. In this study, standard feature scaling is applied before quantum embedding [21], with hyperparameters set to typical values: $\eta = 0.01$, $\tau = 0.05$, $\lambda = 0.1$, $\gamma = 3.0$, and $T = 200$. For downstream quantum analysis, only features mapped to the first and second zones (Z1 and Z2) of the RCP are retained, ensuring both reduced dimensionality and optimal discriminative power.

Selected ML Models. The KNN and LSTM algorithms are selected in this study (Algorithm 1). KNN is valuable for ransomware classification due to its simplicity, interpretability, and robustness in low-dimensional spaces, while LSTM excels at capturing temporal patterns and dependencies in sequential data, which is essential for modeling evolving network behaviors typical of ransomware attacks [25].

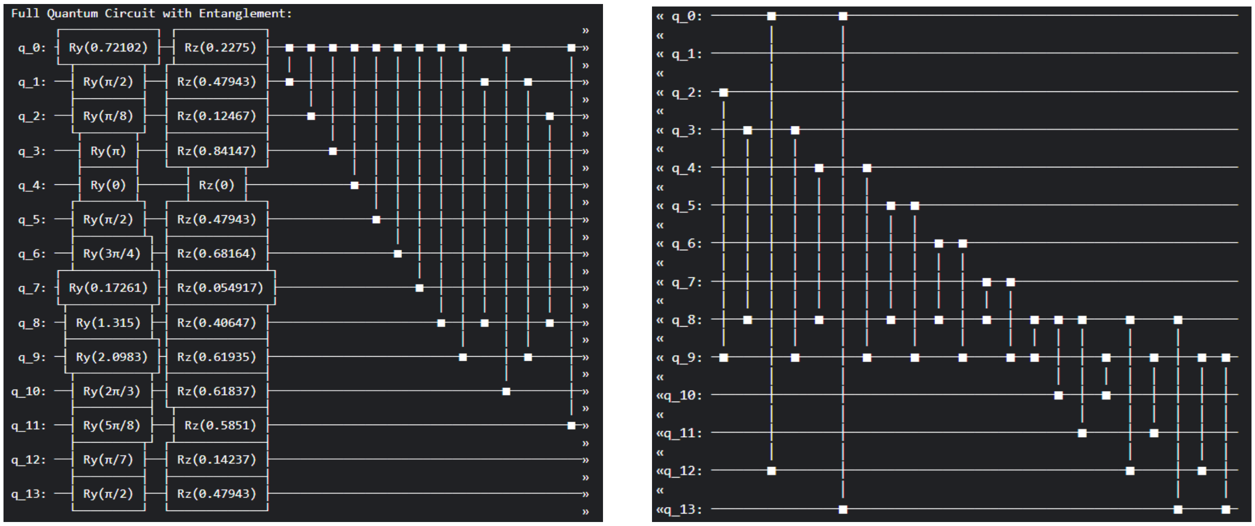
Algorithm 1 Ensemble of KNN and LSTM for Ransomware Classification

Require: Dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, KNN model M_{knn}, LSTM model M_{lstm}
Ensure: Predicted labels $\hat{y}$ for test data
Train M_{knn} on training data
Train M_{lstm} on training data
for each sample x in test dataset **do**
  $p_{knn} \leftarrow M_{knn}.predict(x)$ ▷ KNN predicted label or probability
  $p_{lstm} \leftarrow M_{lstm}.predict(x)$ ▷ LSTM predicted label or probability
  ▷ Combine predictions by majority voting
  if $p_{knn} = p_{lstm}$ **then**
    $\hat{y} \leftarrow p_{knn}$
  else
    $\hat{y} \leftarrow \text{combine}(p_{knn}, p_{lstm})$ ▷ e.g., Weighted average
  end if
end for
return $\hat{y}$

Both the KNN and LSTM models are evaluated using standard classification metrics: the confusion matrix, the Receiver Operating Characteristic (ROC) curve, accuracy (acc), precision (prec), recall (rec), F1-score, False Positive Rate (FPR), and False Negative Rate (FNR) [25]. Algorithm 1 presents the ensemble learning where KNN and LSTM are separately trained. For each test sample, the algorithm obtains predictions from both models. If both agree, it selects that class. If they differ, it uses a combining rule such as weighted average of probabilities or choose the model with higher confidence. Figure 4 provides the quantum circuit representation of the QFM applied in the experimental setup. The quantum encoding circuit developed for the UGRansome dataset effectively maps key features, particularly those related to network behavior and financial impact into a quantum state using RZ rotations. The analysis of rotation parameters reveals that features such as `Netflow/network traffic` (0.61), `BTC` (0.54), `USD` (0.40), and `ExpAddress` (0.68) have comparatively higher RZ rotation angles, indicating a stronger influence on the quantum model's behavior. These features correspond to ransomware traffic and economic loss, which are critical indicators in real-world detection scenarios. In contrast, features with lower rotation values (RZ), such as `Flag` (0.12) and `Clusters` (0.00) (Table 1), contribute less to the entangled quantum state, suggesting a reduced impact on classification performance. This encoding strategy not only enhances the model's ability to capture nonlinear relationships between ransomware characteristics but also provides a basis for feature relevance ranking. When integrated with explainability techniques, this QFM enables interpretable detection decisions and supports forensic analysis of ransomware attacks.

4 Results and Discussions

The feature "ExpAddress" representing the ransomware's cryptocurrency address achieved the highest attribution score (55%), highlighting its critical role in distinguishing ransomware samples based on transactions. The second most influential feature, "Threats" (59%), captures malicious patterns in network traffic, reflecting communication behaviors often associated with ransomware. Its high score underscores the effectiveness of abnormal traffic in detecting anomalies indicative of ransomware [4]. Other top-ranked features include "IPAddress" (40%) associated with the source or target of the traffic and known malicious indicators. This attributes provide contextual insight into the origin of threats and assist in mapping threat actor activity across the network [9]. Conversely, the "Protocol" feature exhibited the lowest scores (Figure 5), indicating limited relevance for effective ransomware detection. The selection of BioSparse-MCP as the feature selection mechanism was motivated by its suitability for handling categorical targets and its efficiency in high-dimensional settings. Compared to alternatives such as PCA and Recursive Feature Elimination (RFE), it offers a better balance between interpretability and computational cost [26]. During the feature selection process, the flag and clusters features were excluded due to their lowest contribution (Figure 5). These low-ranked feature were more likely to introduce noise, and increase the risk of overfitting. Empirical results confirmed that eliminating lower-importance features did not degrade performance, validating that the reduced set captured essential



(a) Quantum encoding circuit.

(b) Full quantum circuit with entanglement

Fig. 4: Quantum circuit representation of 14 UGRansome attributes encoded into a multi-qubit quantum state (Table 1).

patterns necessary for accurate ransomware classification. The analysis of the UGRansome dataset reveals the most influential features identified by the BioSparse-MCP algorithm for enhancing ransomware classification (Figure 5). Prioritising high-impact features thus promotes faster convergence, improved efficiency, and clearer interpretability, which are crucial for practical deployment in cybersecurity applications. Table 3 presents a comparative evaluation of KNN and LSTM, applied to ransomware classification with the aid of Biosparse-MCP selected features. From the results, the KNN demonstrates outstanding performance across all key metrics, achieving an accuracy, precision, recall, and F1 score of approximately 0.9966 (Table 3). This performance suggests that with the proposed feature selection, KNN captures the underlying patterns distinguishing ransomware from benign activities, benefiting from the reduced feature space which mitigates overfitting. In turn, the LSTM model exhibits moderate accuracy (0.6504) with high recall (0.9525). The high recall indicates LSTM's ability to identify ransomware cases correctly, minimizing false negatives, which is critical in cybersecurity contexts where missing an attack could be costly. However, the lower precision (0.6340) and F1 score (0.7613) imply more FPR compared to KNN, potentially due to LSTM's sensitivity in sequence modeling not being fully optimised by the current feature set.

The study investigates the integration of KNN and LSTM into a unified ensemble learning (KNN-LSTM) model (Figure 6).

4.1 KNN–LSTM Ensemble Model Architecture

To evaluate the effectiveness of BioSparse-MCP, the study implemented a parallel ensemble model combining KNN and LSTM classifiers [27]. The ensemble leverages both spatial and temporal characteristics of the selected features, as illustrated in Figure 6.

Model Pipeline. The ensemble pipeline consists of the following stages:

1. **Feature Selection:** BioSparse-MCP selects a sparse and discriminative subset of features from the original network traffic dataset (UGRansome).
2. **Parallel Classification:** The selected features are fed into two independent classifiers:
   – **KNN:** A distance-based classifier using Euclidean distance with $k = 5$ neighbors. It operates on the reduced feature space and benefits from clearer class separation.
   – **LSTM:** A Recurrent Neural Network (RNN) with one LSTM layer (64 units), followed by a dense output layer with sigmoid activation. It captures temporal dependencies in sequential traffic flows.
3. **Ensemble Fusion:** The predictions from KNN and LSTM are combined using soft voting, where probabilistic outputs are averaged to produce the final decision.

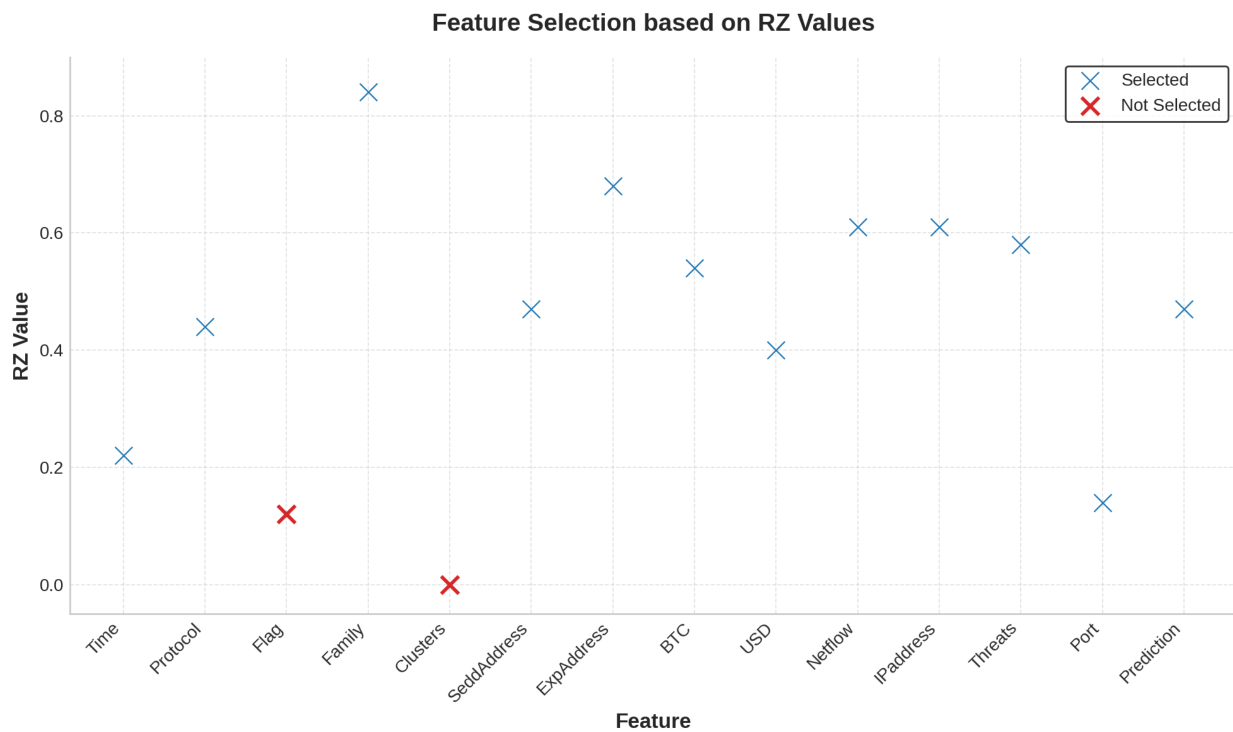


Fig. 5: Selected features based on RZ values.

Python Implementation. The model was implemented using `scikit-learn` and `TensorFlow/Keras`. Key specifications are as follows:

– KNN: `sklearn.neighbors.KNeighborsClassifier(n_neighbors=5, metric='euclidean')`
– LSTM:

```
model = Sequential()
model.add(LSTM(64, input_shape=(timesteps, features)))
model.add(Dense(1, activation='sigmoid'))
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

– `Fusion`: Final prediction computed in Equation 13:

$$\hat{y} = \frac{1}{2}\left(\hat{y}_{\mathrm{KNN}} + \hat{y}_{\mathrm{LSTM}}\right) \tag{13}$$

where $\hat{y}_{\mathrm{KNN}}$ and $\hat{y}_{\mathrm{LSTM}}$ are the predicted probabilities from each model.

Advantages. This ensemble design balances the interpretability and simplicity of KNN with the temporal modeling capacity of LSTM. It is particularly effective when feature selection yields compact, well-separated representations, allowing KNN to perform robustly while LSTM captures sequential ransomware behaviors.

The ensemble learning outperforms both individual models(KNN and LSTM) in terms of precision, recall, and F1 score, demonstrating the effectiveness of combining complementary learning paradigms. However, this performance gain comes at the cost of increased computational time (Table 3). KNN and ensemble models perform very well (Figure 7 and Figure 8).

They keep FPR and missed detections (FNR) very low. LSTM model has much higher FPR and FNR, meaning it is less reliable: many benign samples trigger false alarms, and many ransomware cases are missed. The AUC score of 0.7749 for LSTM suggests fair discriminatory power but leaves room for improvement.

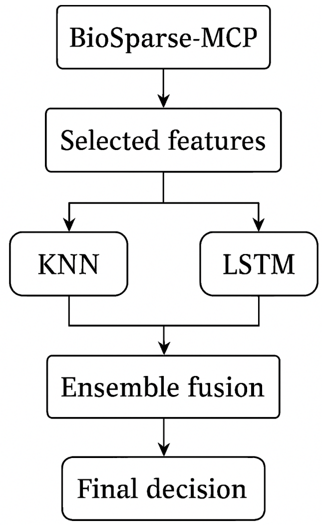


Fig. 6: Pipeline of the KNN–LSTM ensemble model. BioSparse-MCP selects discriminative features, which are then processed in parallel by KNN and LSTM. Their outputs are fused through ensemble voting to produce the final ransomware detection decision.

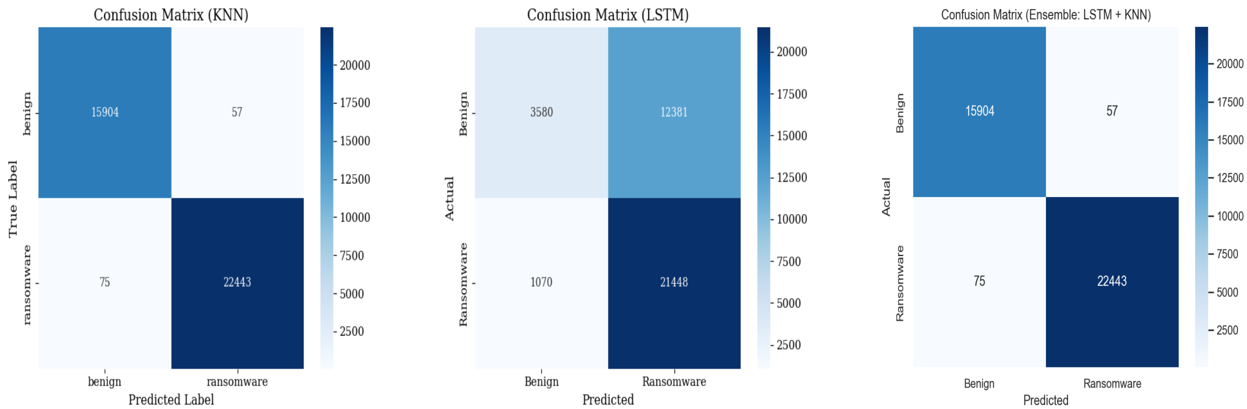
Table 3: Performance Comparison Between Models

Metric	**KNN**	**LSTM**	**Ensemble**
Accuracy	0.9966	0.6504	0.9966
Precision	0.9966	0.6340	0.9975
Recall	0.9966	0.9525	0.9967
F1	0.9966	0.7613	0.9971
AUC	1	0.7749	0.9997
FPR	0.0025	0.3660	0.0025
FNR	0.0047	0.2301	0.0047
Time (seconds)	4.75	6.07	6.21

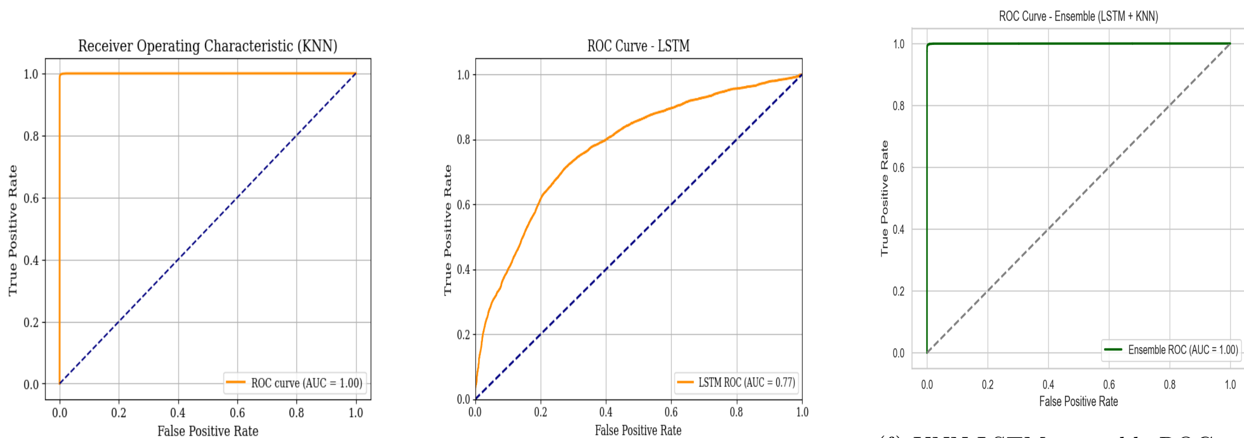
The time taken for model training and evaluation shows that KNN is slightly faster (4.75 seconds) than LSTM (6.07 seconds), which is beneficial for real-time ransomware detection scenarios. Table 4 presents a comparative evaluation of KNN, LSTM, and ensemble learning for selected zones (Z1 and Z2). The Z1 feature subset enhances ransomware classification effectiveness, yielding perfect discrimination and generalising well across different model architectures (Table 4). These results demonstrates that features selected from Z1 via the RCP method yield superior classification performance across all models, achieving perfect scores (F1 = 1.00) for both benign and ransomware classes (Table 4). In contrast, Z2 features result in lower, though still competitive, performance, particularly for the benign class. This performance disparity highlights the effectiveness of spatial partitioning in isolating highly discriminative features, with Z1 capturing the most informative patterns (Table 4). The ensemble model exhibits the greatest robustness, achieving high scores even in Z2 (Figure 8). These results validate the utility of RCP in enhancing model accuracy and interpretability through geometric decomposition of the feature space. Z1 features selected using BioSparse-MCP result in better classification across all models. Performance drops in Z2, especially for the LSTM model, with more FNR, meaning more ransomware samples were misclassified as benign.

4.2 Discussion

Figure 9 presents an interpretable model analysis for the ensemble learning model, which demonstrated the highest classification performance. The explanation supports the quantum circuit analysis by quantifying the contribution of individual features to the model's predictions.



(a) KNN results. (b) LSTM results. (c) KNN-LSTM ensemble results



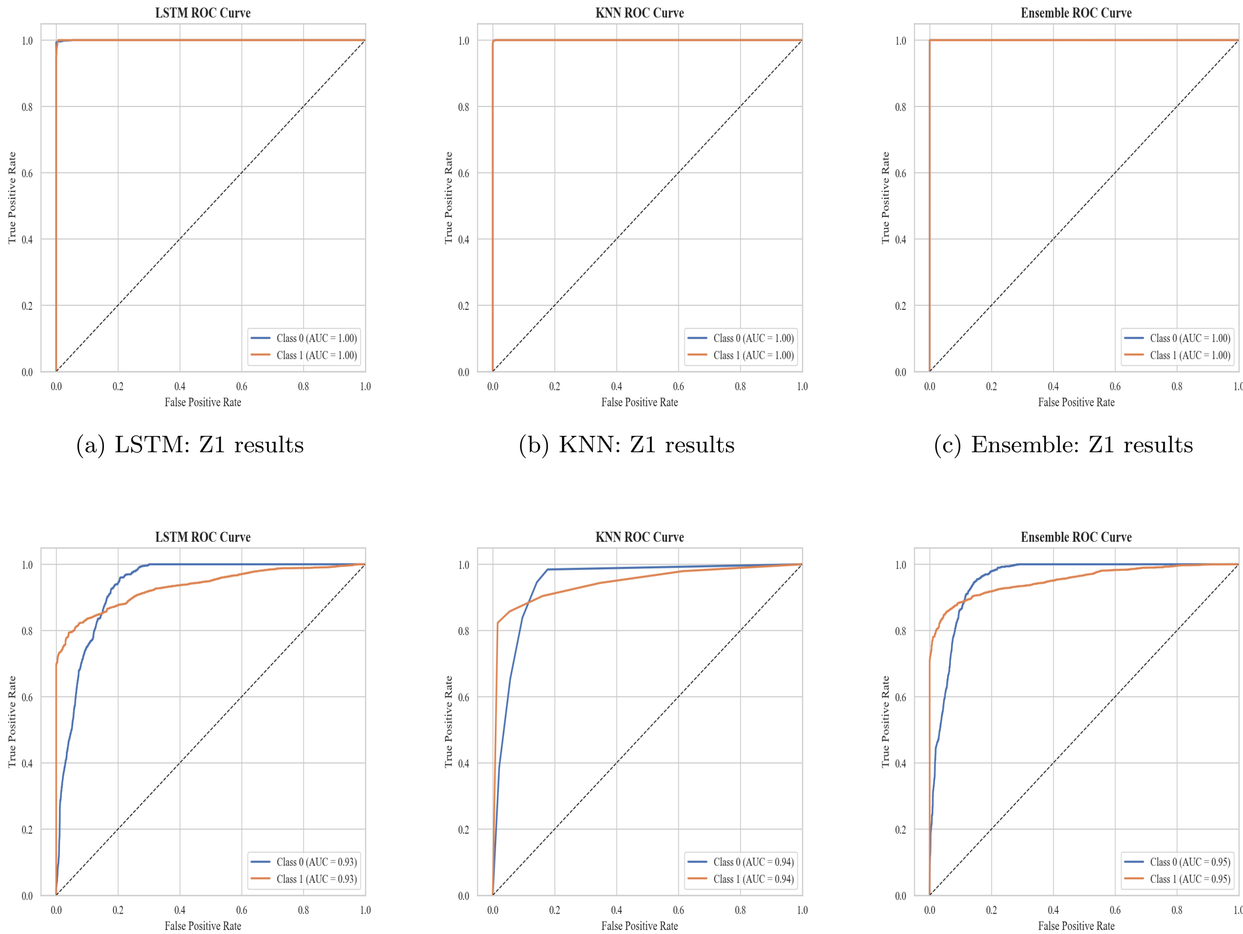
(d) KNN ROC results (e) LSTM ROC results (f) KNN-LSTM ensemble ROC results

Fig. 7: Comparison of confusion matrices and ROC curves for KNN, LSTM, and ensemble models (KNN-LSTM) using selected features.

Table 4: Comparison of Models on Z1 and Z2 Features

Model	**Class**	**Quadrant 1 (Z1)**			**Quadrant 2 (Z2)**		
		Precision	Recall	F1	Precision	Recall	F1
LSTM	Benign	0.99	1.00	1.00	0.73	0.77	0.75
	Ransomware	1.00	1.00	1.00	0.91	0.89	0.90
KNN	Benign	1.00	1.00	1.00	0.77	0.84	0.81
	Ransomware	1.00	1.00	1.00	0.93	0.90	0.92
Ensemble	Benign	1.00	1.00	1.00	0.78	0.86	0.82
	Ransomware	1.00	1.00	1.00	0.94	0.90	0.92

`IPaddress` emerges as the most influential feature in identifying ransomware, reinforcing its high RZ rotation value and the significance of network-level indicators within the quantum encoding. This suggests that specific IP address patterns such as connections to Command-and-Control (C2) servers or recurring access to malicious ranges are key discriminators of ransomware activity. Additionally, features such as `USD` and `Threats`, which also exhibit substantial RZ rotation angles, demonstrate strong interpretability impact. These attributes capture both the financial footprint and known threat signatures associated with ransomware campaigns, aligning with the quantum circuit's prioritisation of economic and behavioral signals. In contrast,



(a) LSTM: Z1 results (b) KNN: Z1 results (c) Ensemble: Z1 results

(d) LSTM: Z2 results (e) KNN: Z2 results (f) Ensemble: Z2 results

Fig. 8: Comparison of ROC curves for KNN, LSTM, and ensemble models (KNN-LSTM) using Z1 and Z2 subsets.

the `BTC` feature shows only marginal influence, which is consistent with its moderate rotation value and may reflect obfuscation techniques employed by attackers or feature sparsity in the dataset. Other moderately weighted features in both the circuit and Local Interpretable Model-agnostic Explanations (LIME) results, including `Time`, `SeedAddress`, `Netflow`, and `ExpAddress` contribute to the overall prediction, further validating the multi-feature entanglement strategy used in the quantum model (Figure 9). The model identifies behavioral anomalies, network attributes, and threat indicators as major predictors of ransomware. While some normal features reduce the probability, the cumulative positive influence of threat related features leads the model to classify the instance as ransomware. These results increase transparency and strengthens trust in ransomware prediction.

4.3 Discussion on temporal modeling

LSTM Temporal Modeling. Although the LSTM model is designed to capture temporal dependencies in sequential data, the temporal ordering in this study was preserved only at the level provided by the original network flow records. No explicit optimisation of sequence length, temporal windowing, or architectural depth was performed [28]. The LSTM architecture and hyperparameters were intentionally kept shallow and fixed to prioritise comparative evaluation rather than exhaustive temporal modelling. As a result, the LSTM may not fully exploit long-range temporal dependencies inherent in ransomware communication patterns.

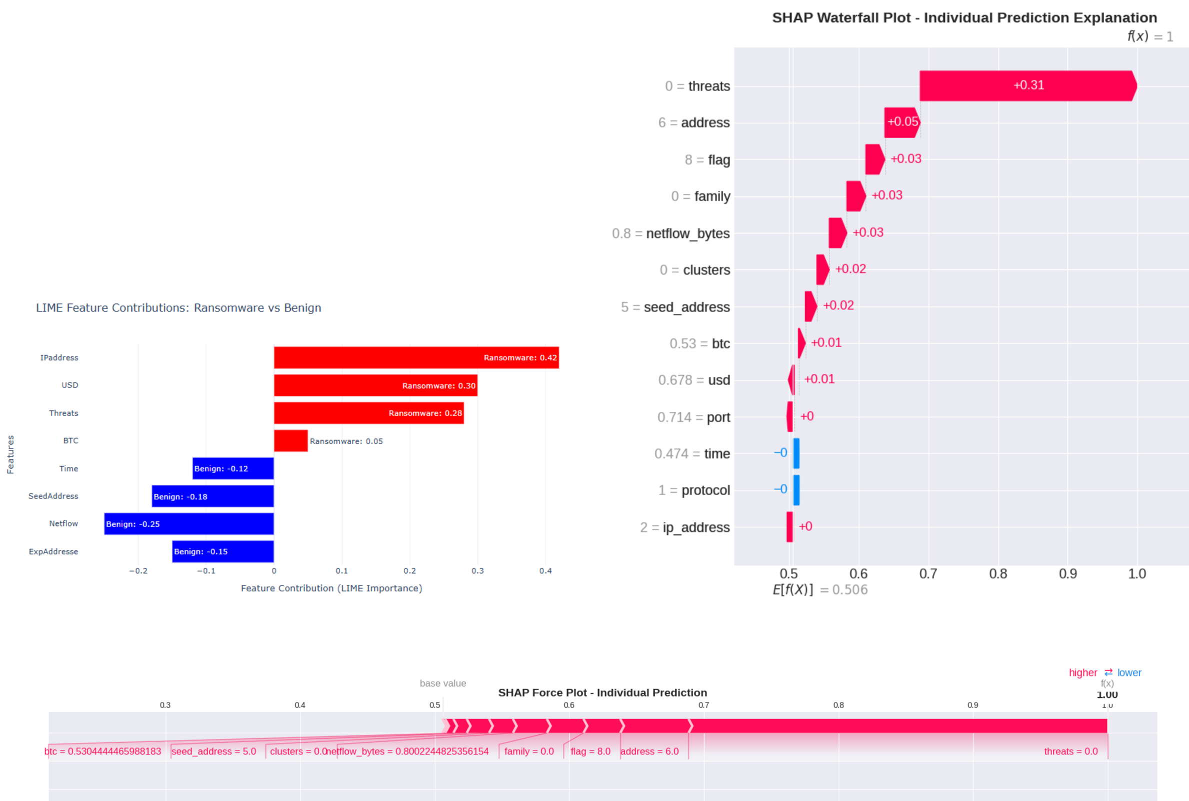


Fig. 9: Explanations for ransomware prediction.

This limitation is acknowledged and considered acceptable within the scope of the study, where the primary focus is on feature selection and representation rather than deep temporal optimisation.

Why KNN Performs Well After BioSparse-MCP. The strong performance of KNN can be attributed to the effect of BioSparse-MCP, which substantially reduces feature redundancy and noise while preserving highly discriminative attributes. By projecting the data into a compact and well-separated feature space, the distance-based assumptions of KNN become more valid [29]. In this reduced space, samples belonging to ransomware and benign classes exhibit clearer local neighborhood structures, enabling KNN to achieve high accuracy with minimal model complexity. This result suggests that effective feature selection can outweigh the benefits of deeper models when the underlying class geometry is well structured [28]; [29].

Generalisation to Unseen Ransomware Families. While the reported results demonstrate strong performance on the UGRansome dataset, generalisation to entirely unseen ransomware families cannot be fully guaranteed. The reliance on network-level behavioural features, rather than static signatures, supports better transferability across variants. However, explicit family-wise holdout experiments were not conducted in this study. Consequently, cross-family generalisation remains an open research direction and is identified as a limitation suitable for future investigation.

Safeguards Against Label Leakage. To mitigate label leakage, all feature selection procedures were applied exclusively on the training split, and no target-derived attributes were used during the BioSparse-MCP optimisation process. Features with direct semantic overlap with the class label were excluded or carefully analysed. Furthermore, performance consistency across independent test partitions suggests that the observed gains are not solely attributable to information leakage but to genuine improvements in feature relevance and representation.

Future Works. An important direction for future work is the inclusion of broader comparative baselines. While this study focused on evaluating BioSparse-MCP with KNN and LSTM classifiers, additional experiments against widely used models such as RF, SVM, Convolutional Neural Networks (CNN), and dimensionality reduction techniques like PCA would provide a more comprehensive assessment of the framework's relative strengths and limitations. Such comparisons could further validate the generalisability of BioSparse-MCP across diverse ML paradigms and highlight its advantages in high-dimensional ransomware detection tasks.

5 Conclusion

This study presents BioSparse-MCP, a feature selection algorithm designed to enhance ransomware classification. Integrated within a KNN–LSTM ensemble model, the algorithm demonstrated its effectiveness by improving classification accuracy, reducing FPRs, and enhancing model interpretability. Experimental evaluation on the UGRansome dataset confirmed that prioritising high-impact features such as ransomware cryptocurrency addresses, threat indicators, and IP addresses enabled robust detection performance with minimal degradation upon removing features with low performance. The ensemble model outperformed individual classifiers, validating the complementary strengths of KNN and LSTM approaches for real-time cybersecurity applications.

Code and Data Availability

The dataset and code used in this study is publicly available on Kaggle [30]. All methods described in this study are quantum-inspired and classically simulated. They should not be conflated with QML approaches, which rely on genuine quantum circuits or hardware.

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